Mathematical Anatomy of Trading Systems

Mathematical Trading Systems

A synthesis of state, geometry, control and falsification

A rigorous mathematical course

Companion to Volumes I and II

September 2026

A route through this volume

Read this companion first, and return to it as the other volumes become more detailed. It explains one common argument in ten short chapters: observations become a belief, a belief becomes a forecast, a forecast becomes a distribution of cash gains, and preferences plus constraints turn those possibilities into an executed policy. The account then supplies evidence for testing the argument.

Futures and SPY options are running examples throughout. The companion introduces no third strategy and assumes no course in stochastic calculus, filtering, control or finance. The main route uses small symbolic examples; the optional reference appendices retain the general mathematical formulas.

How to read the mathematics

Read the prose interpretation of an arrow first, then its equation. The examples introduce symbols locally. If a formula in a reference appendix is unfamiliar, return to the main chapter that explains its role; those appendices are intended for repeated consultation after studying the volumes.

The main text introduces each object when its purpose becomes visible. Sections marked *Derivation* work through the algebra behind a result. Technical appendices retain domain restrictions, exceptional cases and longer proofs. A forward reference to an appendix is permission to defer that detail, not a prerequisite for understanding the next paragraph. The notation table is a reference; no symbol in it needs to be memorized first.

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Principal notation: a reference

Symbol	Role
$\mathcal{G}_t$	Information released by the current decision.
x_t, b_t	State and conditional belief about what is hidden.
$\mathbb{P}, \mathbb{Q}$	Physical probability and a selected pricing probability.
P_t	Conditional law of the future path needed for the trade.
Π	Financed policy gain, including the specified costs and events.
q, h	Contract holdings and stock shares.
$\mathcal{A}(s), \pi_t(s)$	Feasible actions and the selected policy.
W_t, V_t	Account wealth and continuation value; different quantities.

Chapter 1

Observations: what is available when we act?

What problem are we solving?

A trading argument begins with information, not with an optimizer. We need to know which facts the decision may use and which quantities are still uncertain.

1.1 Start with the same two running examples

In the futures system, a trader observes settlement quotes for identified contracts, current collateral requirements and received fills. The first question is whether residual movement contains useful information about the next settlement gain.

In the options system, a trader observes stock price, option bid–ask intervals, announced dividends and received assignment notices. The first question is whether a physical forecast of future market conditions improves a financed spread decision. Today's option price is an observation, not a direct observation of tomorrow's volatility.

1.2 The first mathematical abbreviation

Write O_t for an observation and $\mathcal{G}_t$ for all information released by the time of decision t. The rule is simple: any action chosen at t must be a function of that information. A statistic calculated using tomorrow's outcome is unavailable today even if it is attached to today's row in a later data table.

For example, a gain measured over $[t, t + h]$ becomes observed at $t + h$, not at t. A revised dividend estimate becomes new information when it is released. This timing convention is part of the mathematical model.

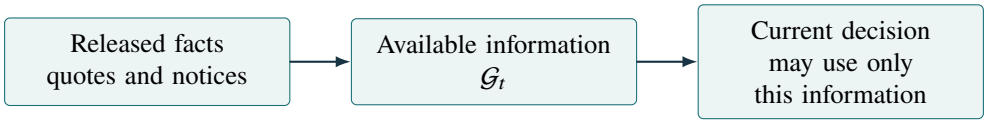


Figure 1.1: The first arrow collects observations by release time; it does not reveal later outcomes.

1.3 Where we are in the argument

The complete route is observations, state or belief, predictive law, gain map, objective, constraints, policy, execution, wealth transition and validation. Each chapter adds one link. At every link ask what comes in, what comes out, and which assumption makes that transformation legitimate. The route is shared; the gains and obligations of futures and options remain different.

What we gained

- Dated observations distinguished from model quantities.
- An information set that prevents future information entering current actions.
- The common route through both trading systems.

Where this goes next

The full observation history can be enormous. The next chapter asks what summary of that history is sufficient for the chosen model.

Chapter 2

State and belief: what should we retain?

What problem are we solving?

We cannot carry an unstructured list of all past observations into every equation. A state retains the information a model needs; a belief retains uncertainty about parts of that state we cannot observe.

2.1 Begin with an observed state

Suppose tomorrow's residual futures gain depends only on today's residual e_t. We can use $x_t = e_t$ as a state. If two lags matter, use $x_t = (e_t, e_{t-1})$. The choice says what predictive information we think can be discarded. It is a hypothesis to test, not an automatic compression.

For options, a constant-volatility price may need only time and stock price. A stochastic-volatility model may need current variance as well. A history-dependent model may need an entire forward-variance curve. More complicated state spaces are introduced only when a simpler state cannot supply the required future law.

2.2 A hidden state requires a distribution

Suppose a latent signal is either high or low, and its prior high probability is p. An observation y has likelihoods $\ell_H(y), \ell_L(y)$. Bayes' rule gives the new high probability

$$p' = \frac{p\ell_H(y)}{p\ell_H(y) + (1-p)\ell_L(y)}.$$

The observation favors a state to the extent that state explains it. This probability, rather than a forced high/low classification, is a belief. For a continuous state we carry a density or probability measure instead. Write b_t for that conditional distribution given $\mathcal{G}_t$.

In the futures example, a noisy residual can update an unobserved expected-return signal. In the options example, prices and variance proxies can update an unobserved variance. Neither reading has to equal the hidden state.

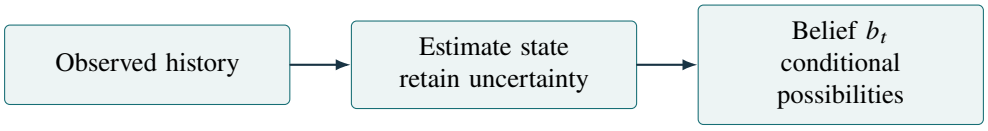


Figure 2.1: A belief is knowledge about a market state. Updating the belief does not change the past market state.

2.3 Parameters are added after the state

A persistence coefficient may also be unknown. Then retain probabilities over that coefficient as well as over the current signal. A model label is a further layer identifying alternative mechanisms. These are three distinct questions: where the state is, how it moves, and which law describes that movement. Appendix A collects the general integral recursion.

What we gained

- An observed state understood as a proposed history summary.
- A hidden-state belief derived from a finite Bayes calculation.
- State, parameter and model uncertainty introduced separately.

Where this goes next

The belief describes what may be true now. Dynamics turn those possibilities into probabilities for what happens next.

Chapter 3

Predictive law: what can happen next?

What problem are we solving?

A signal estimate is not a future outcome. We must combine its present uncertainty with new shocks and preserve dependence between successive dates.

3.1 Propagate a scalar state first

Consider $x_{t+1} = ax_t + w_{t+1}$, where a is known and w_{t+1} is independent of present information, centered, with variance q. If the current belief has mean m and variance p, then

$$\mathbb{E}[x_{t+1} \mid \mathcal{G}_t] = am, \qquad \mathrm{Var}(x_{t+1} \mid \mathcal{G}_t) = a^2 p + q.$$

The first variance term propagates what we do not know now; the second adds tomorrow's innovation. A point estimate would discard the first. For Gaussian beliefs and shocks these two moments specify the full Gaussian forecast. Without that assumption, a mean and variance do not determine the law.

3.2 Unknown coefficients add disagreement

Suppose coefficient choices have posterior weights ω_j, conditional forecast means m_j and variances s_j^2. The forecast is a mixture, with mean $\bar{m} = \sum_j \omega_j m_j$ and variance

$$\sum_j \omega_j s_j^2 + \sum_j \omega_j (m_j - \bar{m})^2.$$

The second term measures disagreement. It appears by subtracting and adding the component mean inside the squared error. A decision should not acquire false confidence merely because an estimator reports one coefficient.

3.3 The two systems forecast different objects

For futures, the ultimate target is the joint path of contract settlement gains. Factor and residual forecasts must be recombined into those coordinates. For options, the target includes stock and variance/surface evolution, as well as contract events needed by the holding rule. Future pricing conditions must be forecast under physical probability, denoted $\mathbb{P}$.

Write P_t for the conditional law of the relevant future path. This symbol abbreviates the preceding construction; it does not specify that construction on its own. A path law contains how dates co-move, not just one marginal distribution for each date.

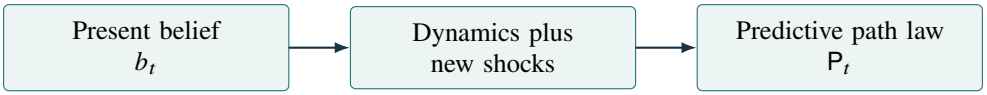


Figure 3.1: The arrow averages conditional transitions over present uncertainty; it does not turn a posterior mean into a certain state.

What we gained

- Propagation of present uncertainty and future innovations.
- Predictive mixtures retaining parameter disagreement.
- A physical path law distinguished from a list of marginal forecasts.

Where this goes next

A future path does not say what our account earns. That requires a holding rule and a gain map appropriate to the contracts.

Chapter 4

Gain map: how outcomes become money

What problem are we solving?

Two portfolios can face the same market path and earn different cash gains. We need an explicit map from the realized path and actual holdings to the financed result.

4.1 The futures map is linear in held-contract changes

For one identified futures contract, let q be the signed holding, m its cash multiplier and F_t, F_{t+1} its settlement quotes. The market gain is $qm(F_{t+1} - F_t)$. Several contracts contribute a sum of these terms. A maturity splice in a chart is not the gain of a held contract.

4.2 An option map also needs future valuation

For one option held without exercise from t to u, let C_t be entry value, C_u exit value and $R_{t,u}$ the cash growth factor. Its financed gain before other events is $C_u - R_{t,u}C_t - K$, where K contains costs in final cash units. A stock hedge contributes its financed stock gains. Exercise changes options into stock and strike cash, so the complete gain must follow that event too.

4.3 Where pricing probability enters

At a future market state, a pricing model maps that state to an option value. Its pricing probability is denoted $\mathbb{Q}$. The physical law $\mathbb{P}$ forecasts which future states occur; the pricing map values a claim at each such state. In a finite illustration, if state z_i occurs with physical probability p_i, the expected exit mark is $\sum_i p_i C(u, z_i)$. Each $C(u, z_i)$ may itself be calculated by a pricing expectation, but its outer weight is still p_i.

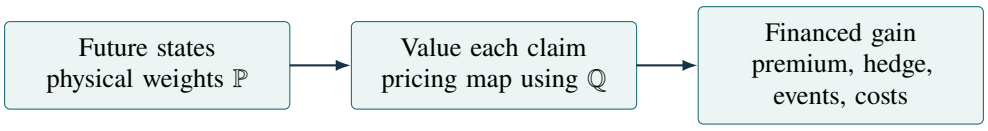


Figure 4.1: The two probabilities enter different operations. Current calibration does not replace the outer physical forecast.

4.4 A common notation only after the examples

Write $\Pi(a, \xi)$ for the gain from action or holding rule a on a future path ξ, with the current account understood. Combining it with P_t gives a distribution of cash gain for every proposed action. This is the input to a decision, not yet the decision itself.

What we gained

- A contract-specific mapping from future paths to cash gains.
- Physical forecasting separated from future option pricing.
- A conditional gain distribution for each proposed holding rule.

Where this goes next

Different gain distributions have different means, tails and funding demands. An objective is needed to say which tradeoffs the investor prefers.

Chapter 5

Objective: what counts as a better decision?

What problem are we solving?

A distribution contains possibilities, not preferences. The investor must say how expected gain, uncertainty, costs and competing uses of capital should be compared.

5.1 Derive the simplest size decision

Let a unit gain have conditional mean μ and variance $s^2 > 0$. Choose size w by maximizing

$$J(w) = w\mu - \frac{\gamma}{2}w^2 s^2, \qquad \gamma > 0.$$

The derivative is $\mu - \gamma w s^2$, giving $w^* = \mu/(\gamma s^2)$. The negative second derivative proves maximality. The forecast supplies μ, s^2; the investor supplies the risk-aversion parameter γ. Two investors can agree on the forecast and choose different sizes.

5.2 Why an objective may need more than variance

Expected utility averages $U(\Pi)$ for a specified increasing utility function U. A tail criterion focuses on severe losses; a cash reserve restriction concerns the ability to meet obligations before the horizon. These are different questions. A bounded expiry spread payoff, for example, does not by itself guarantee that a one-leg assignment is financeable.

For both running examples, costs belong in the cash gain. A separate penalty for concentrated risk expresses a preference and is not another cash expense. That distinction will matter again in the wealth ledger.

5.3 How uncertainty can remove a trade

If the mean lies within ρ of $\widehat{\mu}$, the worst expected gain is $w\widehat{\mu} - \rho|w|$: use the low endpoint for a long holding and the high endpoint for a short holding. With cost $k|w|$, $k \geq 0$, from a zero holding, there is no beneficial position in the scalar quadratic model when $|\widehat{\mu}| \leq \rho + k$. An explicit uncertainty set can therefore justify abstaining rather than simply taking a smaller version of every trade.

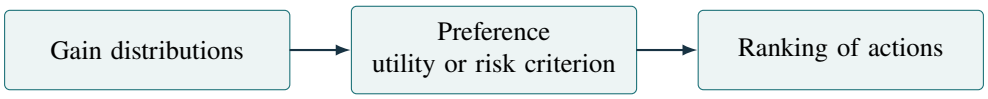


Figure 5.1: The objective ranks possibilities. It does not determine which actions can actually be financed or executed.

What we gained

- A position size derived from a stated risk preference.
- Expected utility, variance and funding concerns distinguished.
- A simple robust no-position threshold.

Where this goes next

The best unconstrained action can be unavailable. The next chapter defines the feasible set and explains what hedges remove.

Chapter 6

Constraints: which bets remain possible?

What problem are we solving?

An objective may favor a position that violates a hedge, capital or contract restriction. We restrict the choices before interpreting any optimum.

6.1 A hedge removes one direction

For three futures exposures, let $\beta_1, \beta_2, \beta_3$ be loadings on one common shock. Neutrality requires $\beta_1 w_1 + \beta_2 w_2 + \beta_3 w_3 = 0$. If $\beta_3 \neq 0$,

$$w_3 = -\frac{\beta_1 w_1 + \beta_2 w_2}{\beta_3}.$$

Two holdings remain free. Their feasible plane is the nullspace of the loading row. A residual forecast is useful only if some remaining feasible direction expresses it at acceptable risk and cost.

For two options with contract deltas d_1, d_2, a stock holding $h = -d_1 q_1 - d_2 q_2$ removes local delta. The option quantities still determine gamma, volatility exposure and exercise obligations. Delta cancellation does not cancel those other responses.

6.2 Why the analogy has limits

Both examples solve linear exposure equations. Futures gains are linear in the stated settlement coordinates; option Greeks approximate a nonlinear pricing map locally. The general matrix/nullspace formulation in Appendix B records this shared algebra and its limits. PCA identifies common variation, but supplies no forecast by itself. Greek geometry identifies local controllable exposures, but supplies no alpha.

6.3 The other restrictions

Let $\mathcal{A}(s)$ denote the feasible actions at current state s. It can impose integer contracts, individual capacity, margin, free-cash reserves and stress limits as well as hedge equations. Bounds depend on the current account. If no action satisfies all restrictions, an optimizer cannot manufacture one; the system needs a declared fallback or liquidation rule.

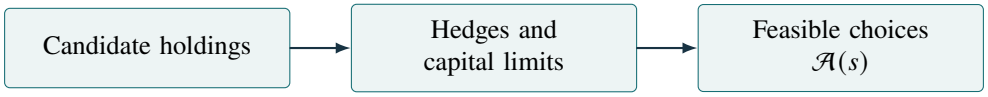


Figure 6.1: Hedging and financing restrict the set on which the objective is optimized.

What we gained

- A hedge understood as a constraint leaving explicit freedoms.
- The shared linear algebra and different economic scope of factor and Greek neutrality.
- A state-dependent feasible action set.

Where this goes next

Feasibility and preferences define a current choice. A policy must also consider how that choice changes the next account state and its future choices.

Chapter 7

Policy: choosing now with tomorrow in view

What problem are we solving?

A position today becomes incoming inventory tomorrow. Reversing it may cost money, and an assignment can consume cash. We need a sequence of decisions that respects this dependence.

7.1 Backward reasoning with two decisions

Let s_1 denote tomorrow's information and account state. If tomorrow is the last decision date, define

$$V_1(s_1) = \max_{a_1 \in \mathcal{A}_1(s_1)} \mathbb{E}[R_1 \mid s_1, a_1],$$

where R_1 includes the declared last-period reward and terminal treatment. At the first date choose

$$V_0(s_0) = \max_{a_0 \in \mathcal{A}_0(s_0)} \mathbb{E}[R_0 + V_1(s_1) \mid s_0, a_0].$$

The expectation averages the random next state. Tomorrow's maximizing action depends on that state, so it can use tomorrow's information without revealing it to today's decision. This is the Bellman principle in its smallest useful form.

Key idea

If I act now, I receive X, move to a random future state, and then behave optimally from there. X is the current reward; the continuation is the value function at the next state.

7.2 What a sufficient trading state contains

For futures, retain the signal belief, actual contracts, cash and collateral. For options, retain the variance/surface belief, options, stock, cash and pending delivery. If drawdown matters, also retain the running wealth maximum. If rough volatility requires a curve, replacing that curve by current variance alone changes the model's conditional predictions.

A policy $\pi_t(s)$ maps the state to an admissible instruction. It can be deterministic while its input is a probability distribution. Averaging over unknown states does not require choosing a random trade.

A receding-horizon policy approximates the future value, executes its first instruction and replans after receiving the next observation.

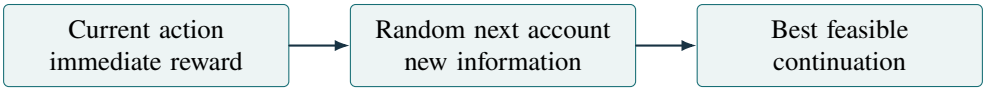


Figure 7.1: The current choice affects both the immediate reward and future opportunities.

What we gained

- A two-period derivation of continuation value.
- A policy distinguished from a repeated static target.
- Inventory, belief and pending obligations included in the state.

Where this goes next

The policy issues an instruction. Execution determines which part of that instruction becomes a real holding.

Chapter 8

Execution: from instructions to fills

What problem are we solving?

The position selected on paper may not be reached immediately or at the assumed price. The gain map must use actual fills, including the positions held while a trade is incomplete.

8.1 A target is not a holding

Let q^- be incoming inventory and q^* the desired target. The instruction requests a change toward q^*. If the actual fill is a, the resulting inventory is $q^+ = q^- + a$. The equality $q^+ = q^*$ is a full-fill assumption, not an identity.

For futures, trading rapidly can increase impact; trading slowly leaves the intended exposure unfilled while prices move. An execution schedule balances these two effects. Its cash cost and its risk penalty are different objects.

For an option spread, one leg can fill before another. Even if the final target cancels delta, the intermediate holdings can have a large delta. A multi-leg order changes the fill mechanism but does not make its price or fill probability certain.

8.2 The execution law belongs in prediction

If fills depend on market conditions, the distribution of realized gain must include that dependence. A favorable price move may be precisely the state in which an order fails to fill. Modeling market gains independently of fills can therefore misstate both expected profit and risk.

The full system specifies an instruction-to-fill law and records actual paid prices. Once a paid price includes spread or impact, that same component is not charged again under another name. Targets help choose instructions; fills determine the holdings that enter accounting.

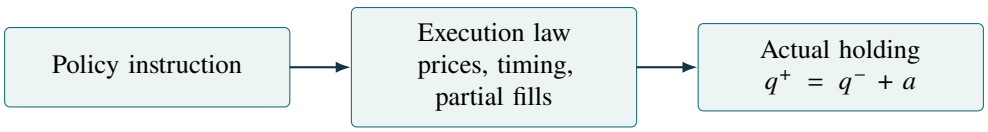


Figure 8.1: Only filled inventory earns subsequent market increments. Execution uncertainty is part of the conditional experiment.

What we gained

- Target, instruction and fill distinguished.
- The cost–waiting-risk tradeoff and temporary option-leg imbalance.
- Execution dependence included in the gain law.

Where this goes next

We can now update the account from realized events. The next chapter checks that the wealth transition balances independently of the forecast model.

Chapter 9

Wealth transition: make the account balance

What problem are we solving?

A statistical explanation of a gain can be wrong while the cash identity is still exact. We derive the wealth update before labeling its pieces as factor profit, volatility profit or alpha.

9.1 Futures: cash and collateral

For one interval, let W_t be equity, K_t trading cost, R_t a common gross cash/collateral growth factor and H_t additional funding cost. With cash-gain vector g_{t+1} per contract and filled holdings q_t^+,

$$W_{t+1} = R_t(W_t - K_t) + (q_t^+)^\top g_{t+1} - H_t.$$

The vector g is multiplier times held-contract settlement changes. If cash and collateral earn different rates, accrue them separately. Transferring cash between those accounts leaves total equity unchanged at the transfer time.

9.2 Options: premium, marks and delivery

With cash X, shares h, stock price S, option counts q and contract marks C, equity is $W = X + hS + q^\top C$. Buying one option at its mark decreases X and increases option assets by the same amount. Exercising a long call replaces that option with shares and a strike-cash payment. A short call assignment reverses those transfers. Pending delivery is recorded once when recognized and transferred, without a second gain, when settled.

9.3 Attribution is a partition of the total

A factor/residual split partitions futures market gains. A Greek expansion explains option mark changes. Neither creates a new term to add to already counted gain. Cross effects can have several attribution conventions while the total wealth change remains fixed. The detailed conservation equations are in Appendix C and the two main volumes.

With no positions, no costs and all cash earning the benchmark, excess gain must be zero. This small check is more fundamental than any estimated alpha: a system that fails it has not defined its gain correctly.

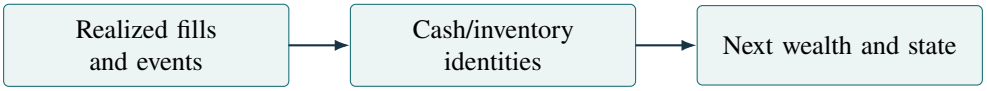


Figure 9.1: The wealth transition closes the decision loop and supplies observations for the next date.

What we gained

- Separate futures and options equity transitions.
- Collateral, premium and delivery transfers distinguished from profit.
- Attribution constrained to reconcile with exact cash gain.

Where this goes next

Balanced accounting supplies the outcome to test. The last main chapter asks whether the forecast improved that outcome and what evidence would reject the claim.

Chapter 10

Validation: can the argument survive new observations?

What problem are we solving?

A complete symbolic system can still be empirically false or economically useless. We specify how future observations test its separate assumptions and its final policy claim.

10.1 First test the arrow that makes the claim

If residual information is proposed to predict futures gains, test future gain distributions beyond a benchmark without that information. If a volatility state is proposed to help an option spread, test future observable prices or variance proxies and the resulting financed policy gains. A low historical reconstruction error or a close present quote fit does not answer either question.

10.2 Then compare complete policies

Let D_t be a predeclared policy gain minus a benchmark gain, with matched capital, costs and obligations. The claim might be $\mathbb{E}D_t > \delta$, where $\delta \geq 0$ is the minimum economically meaningful improvement. A positive sample average is not sufficient evidence. The sampling uncertainty must reflect dependence between dates and contracts.

For example, two overlapping holding periods share market shocks. Treating them as independent underestimates uncertainty. Searching many candidate residuals or spread supports also makes the winning result look better by chance. A valid evaluation fixes the selection procedure and preserves future observations for assessment of that procedure.

10.3 A roadmap to return to

When the result disappoints, locate the failing link. Was the information unavailable in real time? Was the state insufficient? Was the path law poorly calibrated? Did the gain map omit financing or assignment? Did constraints or execution prevent the intended exposure? These questions are more useful than replacing the entire system with a more elaborate model without diagnosis.

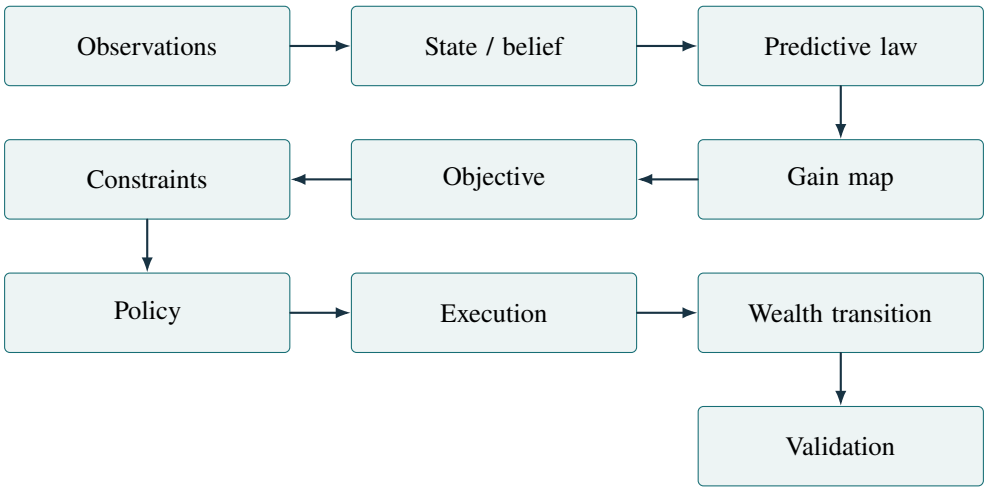


Figure 10.1: The complete roadmap, now read with a meaning for every arrow. Objective and constraints jointly determine a policy; wealth and notices become next-date observations, while validation evaluates the declared experiment.

Mathematical checkpoint

For either running example, explain each arrow aloud before writing its formula. Then name one assumption on that arrow and one future observation that could contradict it. A missing arrow is a missing part of the trading argument.

What we gained

- Predictive fit and economic policy improvement evaluated separately.
- Chronology, dependence and selection included in the experiment.
- A complete roadmap for diagnosing either system.

Where this goes next

The roadmap is complete. The optional reference appendices give compact general formulas for readers returning from the main volumes; they are not prerequisites for the ten-chapter argument.

Appendix A

Reference: probability, pricing and uncertainty

What problem are we solving?

The finite examples in the roadmap now have a general notation. This reference collects the conditional kernels, pricing functionals and uncertainty formulas after their purposes have been explained.

A.1 From observations to a belief

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ be a filtered probability space under the physical law of realized outcomes, and let $\mathcal{G}_t \subseteq \mathcal{F}_t$ contain observations actually released by decision time t. Let $x_t \in \mathcal{X}$ denote an unobserved market state and $\vartheta \in \Theta$ an unknown parameter. A belief is the conditional probability measure

$$b_t(\mathrm{d}x, \mathrm{d}\vartheta) = \mathbb{P}(x_t \in \mathrm{d}x, \vartheta \in \mathrm{d}\vartheta \mid \mathcal{G}_t). \tag{A.1}$$

An observation is a realized quote, settlement, dividend announcement or account event. A latent state is a model object. An estimate is a map from past observations into a point or distribution. A prediction concerns a future random variable. These types must not be exchanged silently.

If $K_\vartheta(x, \mathrm{d}x')$ is the state transition and $\ell_\vartheta(y \mid x')$ is an observation density, prediction and updating obey

$$b^-_{t+1}(\mathrm{d}x', \mathrm{d}\vartheta) = \int K_\vartheta(x, \mathrm{d}x') b_t(\mathrm{d}x, \mathrm{d}\vartheta), \tag{A.2}$$

$$b_{t+1}(\mathrm{d}x', \mathrm{d}\vartheta) = \frac{\ell_\vartheta(y_{t+1} \mid x') b^-_{t+1}(\mathrm{d}x', \mathrm{d}\vartheta)}{\int \ell_{\vartheta'}(y_{t+1} \mid z) b^-_{t+1}(\mathrm{d}z, \mathrm{d}\vartheta')}. \tag{A.3}$$

The denominator must be positive and finite. Linear Gaussian assumptions give Kalman filtering [7]; other models retain the integral recursion even when no finite Gaussian summary is exact.

A.2 A forecast is a law, not an action

For future market path ξ, the forecast is a probability measure $\mathsf{P}_t(\mathrm{d}\xi) = \mathcal{L}_\mathbb{P}(\xi \mid \mathcal{G}_t)$. A policy gain is a separate map $\Pi(s, a, \xi)$. Expected gain is $\int \Pi(s, a, \xi) \mathsf{P}_t(\mathrm{d}\xi)$, while expected utility is $\int U(\Pi(s, a, \xi)) \mathsf{P}_t(\mathrm{d}\xi)$. These generally select different actions.

Futures forecasts enter settlement gains directly after normalization. Options additionally require a pricing functional

$$C_j^M(t,z) = m_j \sup_{\tau \in \mathcal{T}_{t,T_j}} \mathbb{E}_{t,z}^{\mathbb{Q}^M}\left[\frac{B_t}{B_\tau} g_j(S_\tau)\right]. \tag{A.4}$$

The multiplier m_j, payoff g_j, stopping set $\mathcal{T}_{t,T_j}$ and selected pricing law $\mathbb{Q}^M$ define the actual claim. The physical law forecasts future z, then this map supplies future marks. A present calibration cannot replace the missing physical transition.

A $\mathbb{P}$-versus-$\mathbb{Q}$ difference can reflect compensation for risk. For a European auxiliary payoff H and deterministic discount D,

$$V = D\,\mathbb{E}^{\mathbb{P}} H + \mathrm{Cov}^{\mathbb{P}}(M, H),$$

where M is the stochastic pricing kernel with $\mathbb{E}^{\mathbb{P}} M = D$. Thus expected financed payoff is not the same as a no-arbitrage mispricing. American exercise introduces an additional stopping and delivery structure.

A.3 Process, parameter and model uncertainty

For model M and parameter ϑ, total predictive variance obeys

$$\mathrm{Var}(\Pi \mid \mathcal{G}_t) = \mathbb{E}[\mathrm{Var}(\Pi \mid \mathcal{G}_t, M, \vartheta) \mid \mathcal{G}_t] + \mathrm{Var}(\mathbb{E}[\Pi \mid \mathcal{G}_t, M, \vartheta] \mid \mathcal{G}_t). \tag{A.5}$$

The first term concerns shocks conditional on a model; the second concerns uncertain forecasts across parameters and models. A plug-in covariance can omit the second even when it drives the recommended position.

A Bayesian rule averages using specified posterior probabilities. A robust rule uses $\sup_a \inf_{P \in \mathcal{U}} \mathbb{E}_P[U(\Pi^a)]$. These need not agree. For an ellipsoidal mean set $\mu = \widehat{\mu} + Lu,\ \|u\| \le \rho$, Cauchy–Schwarz gives

$$\inf_{\mu} w^\top \mu = w^\top \widehat{\mu} - \rho \|L^\top w\|.$$

The resulting norm penalty is a concrete uncertainty statement [5]. It can eliminate a trade rather than merely reduce its size.

Randomization is a fourth object. If continuation value $Q(s,a)$ is already specified, then $\int Q(s,a)\pi(\mathrm{d}a \mid s) \le \sup_a Q(s,a)$. Belief uncertainty therefore does not by itself imply a random action. Entropy regularization changes the criterion; exploration is valuable only if an action changes information or future opportunities in a modeled way.

What we gained

- General Bayes prediction and updating.
- The physical-law/pricing-functional composition.
- Total variance, robust objectives and policy randomization distinguished.

Where this goes next

The next reference develops the common linear exposure geometry and states its economic limits.

Appendix B

Reference: factor and Greek geometry

What problem are we solving?

The three-asset and two-option examples become a general linear system. We state the range and nullspace solution and then restore the nonlinear terms that option neutrality leaves behind.

B.1 A shared linear problem

Suppose a linear gain representation is $r = Bf + \varepsilon$, with $r \in \mathbb{R}^n$, $B \in \mathbb{R}^{n\times k}$, $f \in \mathbb{R}^k$. A futures exposure w earns $w^\top r = (B^\top w)^\top f + w^\top \varepsilon$. Hence $B^\top w = 0$ annihilates modeled factor exposure.

For option marks $C(z) \in \mathbb{R}^m$ at risk state $z \in \mathbb{R}^d$, let $G = (\partial C_j/\partial z_i)_{ij} \in \mathbb{R}^{d\times m}$. Then $q^\top \Delta C \approx (Gq)^\top \Delta z$. A first-order hedge sets selected rows of Gq to chosen targets.

Both problems take the form $Ax = b$. If A has rank r, feasible solutions exist exactly when $b \in \operatorname{range} A$. With Moore–Penrose inverse A^+ and an orthonormal nullspace basis N,

$$x = A^+ b + Nz. \tag{B.1}$$

The parameter z describes the remaining $n - r$ freedoms. These are where expected gain, transaction costs, concentration and robustness choose among hedged portfolios.

The distinction appears at the next order. Even if $Gq = 0$,

$$q^\top \Delta C = \frac{1}{2}\Delta z^\top \left(\sum_j q_j \nabla^2 C_j(z)\right)\Delta z + \text{higher-order and time terms.} \tag{B.2}$$

For options, tangent neutrality leaves curvature, jumps and exercise. In the stated futures coordinate model, the gain remains linear in the realized return vector, although estimated factor loadings and unmodeled risks can change.

B.2 Statistical geometry versus economic geometry

PCA solves $\max_{V^\top V = I_k} \operatorname{tr}(V^\top \Sigma V)$, or equivalently minimizes expected squared reconstruction error [6]. It does not maximize forecast utility. A Greek matrix differentiates a pricing map; its singular vectors identify locally controllable risk directions, not profitable trades.

Inverse conditioning matters in both settings. A nearly repeated covariance eigenvalue destabilizes an individual principal direction. A small singular value of a Greek matrix magnifies the quantity

needed to reach an exposure target. A nearly singular covariance exaggerates mean-variance positions. Regularization must be tied to a declared uncertainty or preference, since changing a matrix can change the economic question.

Mathematical checkpoint

If the available neutral subspace has dimension zero, what could a residual forecast accomplish? If it has positive dimension, what additional information is still needed to justify a trade?

What we gained

- The general affine exposure solution.
- Curvature remaining after first-order option neutrality.
- Conditioning risks in projection, hedging and position sizing.

Where this goes next

The next reference puts the two exact ledgers alongside the general Bellman equation.

Appendix C

Reference: cash conservation and control

What problem are we solving?

The roadmap kept the ledgers short. These equations give signed inventory and delivery maps together with a general controlled transition.

C.1 Two ledgers

For futures after settlement, marked account equity is cash plus collateral. With equal gross accrual R_t^f, normalized gain r_{t+1}, scale D_t and actual post-trade contracts q_t^+,

$$W_{t+1} = R_t^f (W_t - K_t) + (q_t^+)^\top D_t r_{t+1} - H_t. \tag{C.1}$$

Here K_t is trading cost and H_t extra funding cost. Contract notional is not purchased as premium, and transferring margin does not destroy equity.

For options, shares and cash, marked equity is $W = X + hS + q^\top C$. A trade (a, b) changes

$$(q, h, X) \longmapsto (q + a, h + b, X - a^\top C - bS - K),$$

so equity falls by K. Signed exercise removal e_j changes

$$q_j \mapsto q_j - e_j, \quad h \mapsto h + \sum_j m_j \chi_j e_j, \quad X \mapsto X - \sum_j m_j \chi_j K_j e_j.$$

Here $\chi_j = 1$ for calls and -1 for puts. This is physical delivery. An option premium is accompanied by an asset or liability, while future exercise can create an intermediate stock position. The American SPY structure is confirmed by the exchange and clearinghouse sources [3, 8].

C.2 Why the target is only an intermediate object

Let s_t contain beliefs, actual inventory, cash and all pending obligations. A sufficient state satisfies $\mathbb{P}(s_{t+1} \in A \mid \mathcal{G}_t, a_t) = K_t(A \mid s_t, a_t)$. For terminal value Φ and flow reward R_t, dynamic programming gives

$$V_T = \Phi, \qquad V_t(s) = \sup_{a \in \mathcal{A}_t(s)} \int [R_t(s, a, s') + V_{t+1}(s')] K_t(\mathrm{d}s' \mid s, a). \tag{C.2}$$

This follows by conditioning on the first transition and selecting an optimal admissible continuation. Attainment needs finite feasible actions or the relevant compactness, measurability and continuity assumptions [4].

Execution makes actual inventory differ from target inventory. Futures execution balances impact against unfilled-inventory risk [1]; options execution also creates temporary Greek imbalances between legs. In both, the ledger uses fills. Multi-period planning must account for their future effects [2].

With terminal utility, cash costs are deducted inside wealth and not again as a separate utility expense. With additive gain rewards, terminal value must not count the same gains twice. A risk penalty is not a cash debit. Robust dynamic programming additionally needs conditional ambiguity sets stable under pasting; otherwise repeated local worst-case choices change the original model-uncertainty problem.

What we gained

- The two self-financing ledgers in full common notation.
- A controlled transition and Bellman value.
- The accounting and time-consistency qualifications for the objective.

Where this goes next

The final reference records the dependence-aware test and the correspondence between the complete systems.

Appendix D

Reference: validation and the complete correspondence

What problem are we solving?

A compact specification is useful after the argument is understood. This final reference records the test assumptions and maps each role in one system to its counterpart in the other.

D.1 The experiment must be allowed to fail

Define training, model-selection and final evaluation periods in chronological order. All transformations and actions at t must be $\mathcal{G}_t$-measurable. Future-horizon labels become available only at the end of their outcome intervals. Contract eligibility, costs, uncertainty sets, risk constraints, benchmarks, candidate families and decision thresholds are fixed before final evaluation.

For a predeclared score or gain improvement D_t, the economic null is $\mathbb{E}D_t \le \delta_{\text{econ}}$, with a symbolic threshold for meaningful improvement. Under an appropriate dependent-data central limit theorem,

$$\sqrt{N}(\overline{D} - \mathbb{E}D_t) \Rightarrow \mathcal{N}(0, \omega^2), \qquad \omega^2 = \sum_{\ell \in \mathbb{Z}} \operatorname{Cov}(D_t, D_{t-\ell}).$$

Stationarity, suitable mixing, moments and a valid long-run variance estimate are required. Overlapping horizons and shared contract shocks invalidate a naive independent-observation count.

Trying many residual constructions or many strike/maturity supports requires a multiple-testing correction that includes the search. The final winner is not a predeclared hypothesis merely because only that result is reported [9]. Structural breaks can invalidate both the forecast law and the test's limiting distribution.

Role	Futures	SPY options
Market observation	Maturity-specific futures settlements, margin and fills	SPY, option intervals, dividends and notices
Statistical state	Factor and residual-return coordinates	Latent variance and admissible surface/history state
Forecast	Conditional normalized futures gain path	Joint spot, surface, dividend and assignment path
Portfolio variable	Dollar exposure w, contract count q^F	Option-leg count q^O, stock shares h
Controlled subspace	Factor-neutral $B^\top w = 0$	Greek constraint $Gq^O + g_S h = b$
Risk	Conditional covariance plus tails, liquidity and drawdown	Nonlinear repricing, surface tails, delivery and funding
Target selection	Costs, margin and robust expected gains	Costs, capital, model ambiguity and assignment
Dynamic state	Belief, inventory, cash and margin	Belief, options, stock, cash and pending delivery
Gain mechanism	Variation margin plus collateral accrual	Premium-funded marks, stock gains and physical exercise
Validation	Residual predictive value and net policy improvement	Physical forecast value and net hedged improvement

Table D.1: A correspondence of roles, not an assertion that the instruments have interchangeable risks.

D.2 The correspondence

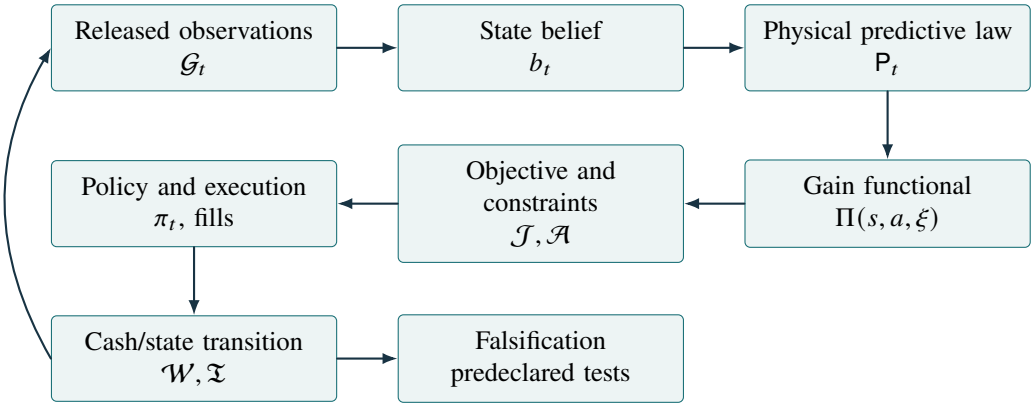


Figure D.1: The repeated architecture. The feedback arrow advances time; it does not revise past information sets.

D.3 From the roadmap to an executable experiment

The two sheets at the end of this document specialize the models already developed in the main volumes. Each sheet gives a dated observation map, estimators, a physical predictive law, a feasible optimization problem, an instruction map and a cash transition. The futures sheet uses observed-state VAR forecasts and robust one-period control; the options sheet uses a Heston posterior, American marks and a finite-action one-period utility controller. These declared approximations need not agree with a full-horizon Bellman policy.

Read each numbered row as a calculation whose output is consumed by a later row. The final equation substitutes those calculations into the next action. Python representations describe the

mathematical objects without prescribing implementation code.

What we gained

- A dependence-aware economic hypothesis test.
- A role-by-role comparison of the futures and options constructions.
- Two compact operational references joining inferred quantities to the next action.

Where this goes next

After the references, use the two closing sheets to reconstruct and inspect an experiment. Return to the main roadmap or the relevant volume whenever a derivation is needed.

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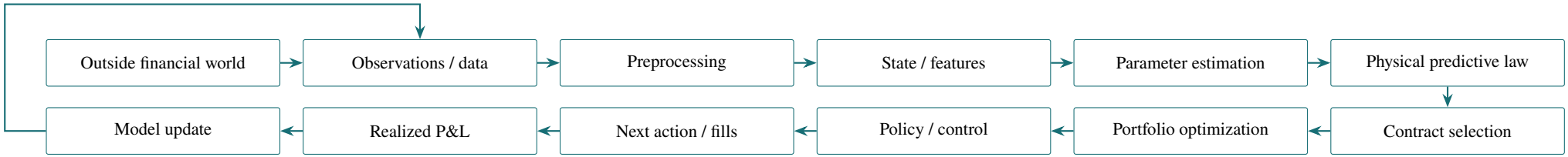
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Multi-asset futures: data, thesis and path law

One-period baseline retained; optional slow thesis state and multi-period control use the same gain and cash maps.



Observations update beliefs; only decision dates (or mandatory risk events) trigger strategic orders. Execution has its own faster clock.

Three clocks. Observations arrive at o_j; decisions occur at t_k with spacing Δ_d; the economic thesis expires at a fixed T^{th} (e.g. weeks). Planning depth H counts decision intervals, $t_{k+H} \le T^{\text{th}}$; it is not the observation count. Below k indexes decisions and $\mathcal{F}_k$ contains only released information. Freeze training length N, PCA rank $r < n$, scales, priors, calendar/surprise laws, risk/cost bounds, seed and validation splits. All money uses one currency. Do not roll the thesis deadline forward after losses.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
F0. Data, gains, coordinates $O_{\le k}$: dated records; $g_s, x_s \in \mathbb{R}^n$. $D \in \mathbb{R}^{n\times n}$ positive diagonal; $\bar r \in \mathbb{R}^n$. $B = [V\ U] \in \mathbb{R}^{n\times n}$ orthogonal, $V \in \mathbb{R}^{n\times r}$.	Released settlements, multipliers m_i, liquidity, funding, margin and fills enter $g_{i,s} = m_i(F_{c_i(s-1),s} - F_{c_i(s-1),s-1})$, using the maturity held over that interval. Never splice a roll price jump into gain. On N training gains, $D_{ii} = \mathrm{sd}(g_i) > 0$, $r_s = D^{-1}g_s$, $\bar r = N^{-1}\sum r_s$. Eigenvectors of $S_r = (N-1)^{-1}\sum(r_s - \bar r)(r_s - \bar r)^\top$ give B, ordered with fixed signs/ties. Fast factor/residual coordinates: $x_s = B^\top(r_s - \bar r)$. Freeze/recompute the whole block in one basis. **To F1–F3, F5.**	NumPy data `(N,n)`, basis/covariance `(n,n)`, state `(n,)`; timestamped contract records.
F1. One-step baseline fit $\widehat A \in \mathbb{R}^{n\times n}$, $\widehat Q \in \mathbb{S}^n_{++}$; $\mu_k \in \mathbb{R}^n$, $\Sigma_k \in \mathbb{S}^n_{++}$. $\widehat\theta$: all fitted coefficients.	$x_{s+1} = Ax_s + \varepsilon_{s+1}$, $\varepsilon \sim \mathcal{N}(0, Q)$. $\widehat A = (\sum x_{s+1}x_s^\top)(\sum x_s x_s^\top + \lambda I)^{-1}$, $\widehat Q = (N-1)^{-1}\sum(x_{s+1} - \widehat A x_s)(x_{s+1} - \widehat A x_s)^\top + \epsilon I$, $\lambda, \epsilon > 0$. Reject spectral radius $\rho(\widehat A) \ge r_* < 1$. Baseline $g_{k+1} \mid \mathcal{F}_k \sim \mathcal{N}(\mu_k, \Sigma_k)$, $\mu_k = D(\bar r + B\widehat A x_k)$, $\Sigma_k = DB\widehat Q B^\top D$. **To F3, F5; this is one step, not a thesis-horizon forecast.**	Arrays `(n,n)`, `(n,)`; fit/transition callables; SciPy Gaussian law.
F2. Slow thesis extension $u_k \in \mathbb{R}$: latent valuation anchor; $\ell_k \in \mathbb{R}$: observed reference-spread level. $d_k = u_k - \ell_k$: remaining gap; $y_k \in \mathbb{R}$: economic proxy. $c_{k+1} \in \mathbb{R}^p$: catalyst surprises; $b_k = \mathrm{Law}(u_k \mid \mathcal{F}_k)$.	Choose a fixed reference basket $w_0 \in \mathbb{R}^n$, capital $A_0 > 0$, and a predeclared economic valuation proxy y in the same units: $\ell_{k+1} = \ell_k + w_0^\top g_{k+1}/A_0$, $u_{k+1} = \bar u + \phi(u_k - \bar u) + \beta^\top c_{k+1} + \sigma_u\eta_{k+1}$, $y_k = u_k + \tau_y\nu_k$, $x_{k+1} = Ax_k + a_d d_k + J_c c_{k+1} + \varepsilon_{k+1}$. Here $a_d \in \mathbb{R}^n, J_c \in \mathbb{R}^{n\times p}, \beta \in \mathbb{R}^p$, $0 < \phi \le 1$, $\sigma_u, \tau_y > 0$; independent standardized Gaussian η, ν. Persistence is estimated ϕ near one and small σ_u, not immunity to evidence. At announced releases $c \sim \mathcal{N}(\bar c, C_c)$, $C_c \in \mathbb{S}^p_+$, fitted to past surprises; otherwise $c = 0$. Filter from a proper prior: $b_{k+1}(u) \propto p(y_{k+1} \mid u)\int p(x_{k+1}, u \mid x_k, u', \ell_k) b_k(u')\,\mathrm{d}u'$; maximize predictive log likelihood (EM) for coefficients. Between decision-grid VAR observations, released y's update the current anchor likelihood; quotes update risk. $a_d = J_c = 0$ recovers F1. **To F3–F6.**	Filter mean/variance `(1,)`, `(1,1)`; coefficients `(n,p)`, `(p,)`. Proxy map and catalyst calendar fixed before testing.
F3. Joint physical path law $Z_k = (x_k, u_k, \ell_k) \in \mathbb{R}^{n+2}$; K^P_θ: F1/F2 transition. $\mathsf{P}^H_k \in \mathcal{P}((\mathbb{R}^{n+2})^{H+1})$. $L_k \in \mathbb{R}^{n\times n}$, $\rho_\mu \ge 0$: mean uncertainty.	$\mathsf{P}^H_k(\mathrm{d}Z_{k:k+H}) = b_k(\mathrm{d}u_k)\prod_{j=k}^{k+H-1} K^P_{\widehat\theta}(\mathrm{d}Z_{j+1} \mid Z_j)$, observed x_k, ℓ_k fixed, catalysts integrated. The endpoint $\mathbb{P}(Z_{k+H} \mid \mathcal{F}_k)$ is only a marginal. For F1 alone it gives $x_{k+H} \sim \mathcal{N}(A^H x_k, \sum_{j=0}^{H-1} A^j Q (A^j)^\top)$; cumulative gain and interim risk need cross-time dependence. Use $\mu_k = \mathbb{E}^P_k g_{k+1}$, $\Sigma_k = \mathrm{Cov}^P_k(g_{k+1})$, including anchor uncertainty. Block-bootstrap refits give $\mu_k^{(b)}$, $L_k L_k^\top = \widehat{\mathrm{Cov}}_b(\mu_k^{(b)}) + \epsilon_\mu I$; ρ_μ is a frozen quantile of $\|L_k^{-1}(\mu_k^{(b)} - \mu_k)\|$. Hold each fitted parameter set fixed per path. **To F5–F6.**	Paths `(N_paths,H+1,n+2)`, gains `(N_paths,H,n)`; Gaussian distribution or simulation callable; bootstrap means `(N_boot,n)`.
F4. Thesis status / expression $p_k = b_k(u_k - \ell_k > \delta) \in [0,1]$; $z_k \in \{0, 1, \dagger\}$: waiting, active, closed. $p_{\text{on}} > p_{\text{off}}, \delta, \delta_{\text{exit}} \ge 0$; eligible mask $j_k \in \{0,1\}^n$.	For a positive-gap thesis, activate at $p_k \ge p_{\text{on}}$; retain it between thresholds. Close on convergence $\mathbb{E}_k d_k \le \delta_{\text{exit}}$, $p_k \le p_{\text{off}}$, released adverse evidence $y < y_{\min}$ or $c_j < \underline{c}_j$, or T^{th}; freeze thresholds and label reasons. Waiting forbids new thesis risk; † cancels entry orders and flattens, and is absorbing until a new thesis record. Risk/margin failures act immediately. The thesis is u, d, b, not a contract ID: choose maturities, spreads, hedges or zero anew. Shares/options require their own pricing/ledger. Eligibility $j_i = 1$ requires valid liquid quotes, capped spreads and dates before notice/roll cutoff; held ineligible legs exit. Disabling this extension uses F1/F5 without the thesis gate. **To F5–F6.**	Scalar probabilities, mode/deadline record, Boolean mask `(n,)`. Log thesis exit separately from temporary zero exposure.

Futures: strategic control, execution and accounting

The baseline uses H=1. Longer planning adds continuation value; observing another tick does not itself authorize another position change.

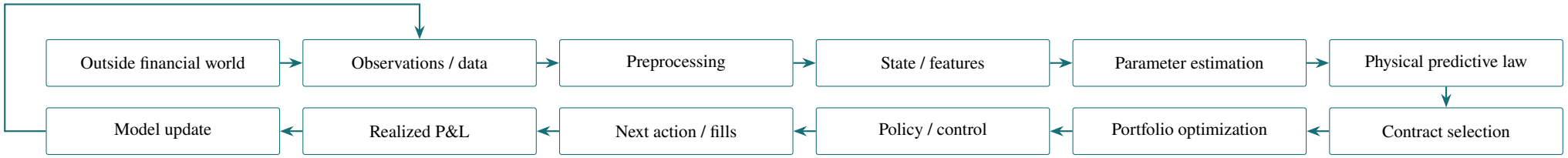
Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
F5. Baseline target / constraints $s^{\text{acct}} = (q^-, X, M) \in \mathbb{Z}^n \times \mathbb{R}^2$, $W = X + M$. Target $q \in \mathbb{Z}^n$, roll map $\mathcal{R}(q) \in \mathbb{Z}^{2n}$. $s, \eta \in \mathbb{R}^{2n}_+$, $\ell^M \in \mathbb{R}^n_+$: spread/impact/margin.	With diagonal roll flags R^{roll}, $a = \mathcal{R}(q) = (-R^{\text{roll}}q^-, q - (I - R^{\text{roll}})q^-)$. Read spreads/margin; fit nonnegative impact from fills. Define $K(a) = s^\top\lvert a\rvert + \sum \eta_i a_i^2$, $M^+(q) = (\ell^M)^\top\lvert q\rvert$, $d^c(q) = (R^M - R^f)M^+(q) - R^f K(a)$, $v(q) = q^\top \Sigma_k q$. Baseline score $J_1(q) = [q^\top\mu_k - \rho_\mu\|L_k^\top q\| + d^c(q)]/A_0 - \gamma v(q)/(2A_0^2)$. Feasible C_k: $\lvert q_i\rvert \le j_i \bar{q}_i$, $\|Dq\|_1 \le L_g$, $\lvert V^\top Dq\rvert \le b$, $\lvert H_g Dq\rvert \le c_g$, $W - K - M^+ \ge R_{\text{cash}}$, stress-equity floors, and $-q^\top\mu + \rho_\mu\|L^\top q\| - d^c + \sqrt{v}\,\varphi(\Phi^{-1}(\alpha))/(1-\alpha) \le L_{\text{ES}}$. This Gaussian ES formula applies only to the one-step Gaussian law; use simulated loss ES for longer paths. **To F6.**	CVXPY integer vector `(n,)` or bounded enumeration; constraints use supplied caps, $H_g \in \mathbb{R}^{p_g \times n}$. Gross factors $R^f, R^M > 0$; normal PDF/CDF φ, Φ.
F6. Bellman / stochastic MPC $s_k = (\mathcal{F}_k$ summary, $b_k, z_k, T^{\text{th}} - t_k, s_k^{\text{acct}}$, pending orders). $V^h, Q^h : \mathcal{S} \to \mathbb{R}$, $\mathcal{S} \times \mathcal{A} \to \mathbb{R}$. $\eta_T, h_0 \ge 0$: turnover/switch preferences.	Let $\Pi = W' - R^f W$ from F7 and $r_F = \Pi/A_0 - \gamma\,\mathrm{Var}_k(\Pi)/(2A_0^2) - \rho_\mu\|L_k^\top q\|/A_0 - \eta_T\|a\|_1$. $Q_k^h(s,a) = \mathbb{E}_k^P[r_F + V_{k+1}^{h-1}(s')]$, $V_k^h(s) = \max_{a \in \mathcal{A}_k(s)}\{Q_k^h(s,a) - h_0 \mathbf{1}_{a \ne 0}\}$. Choose $V^0 = 0$ to recover F5 at $H = 1, \eta_T = h_0 = 0$; use a specified continuation value at a shorter planning boundary, and $-K^{\text{liq}}/A_0$ at the thesis endpoint. Simulate proxy observations and re-filter at future nodes: actions never see latent states or later shocks. Stochastic MPC executes the first action of this nonanticipative tree; full Bellman control uses the remaining horizon. No-action region: $\max_{a \ne 0}[Q^H(s,a) - Q^H(s,0)] \le h_0$, if holding is feasible. F4 hysteresis and this band suppress reversals; mandatory exits/rolls override them. **To F7.**	Scenario tree / value callable; finite belief representation $\varphi_s(s) \in \mathbb{R}^{d_s}$. Costs are already in Π; turnover and switch penalties are preferences, not second cash debits.
F7. Separate execution / ledger $a^{\text{str}} \in \mathbb{Z}^{2n}$: strategic parent order; $y_e, v_e \in \mathbb{R}^{2n}$: remaining order/rate. $\Lambda_e \succ 0, \Omega_e \succeq 0 \in \mathbb{R}^{2n \times 2n}$, $T_e \ll \Delta_d$, $\lambda_e \ge 0$.	Execution chooses $v_e = -\dot{y}_e$ to minimize $\int_0^{T_e}[s_e^\top\lvert v_e\rvert + v_e^\top\Lambda_e v_e + \lambda_e y_e^\top\Omega_e y_e]\,\mathrm{d}u$, $y_e(0) = a^{\text{str}}$, $y_e(T_e) = 0$, subject to lot/rate/interim-risk limits. Fit impact Λ_e from fills and quote-change covariance Ω_e; carry unfilled quantities if completion fails. Feed expected cash cost back into F5/F6, replacing its cost estimate; execution-risk penalty is separate. Full immediate fills remain the baseline: $q^+ = q^- + \tilde{a}^{\text{old}} + \tilde{a}^{\text{new}}$, $X^+ = X + M - M^+(q^+) - K(\tilde{a})$, $W' = R^f X^+ + R^M M^+(q^+) + (q^+)^\top g'$, $\Pi = W' - R^f W$. Reset $M' = (\ell^{M'})^\top\lvert q^+\rvert$, $X' = W' - M'$. With gradual fills, accrue cash event by event and use $\int q(u)^\top\,\mathrm{diag}(m)\,\mathrm{d}F(u)$ over actual maturities. **To next account/filter.**	Execution optimizer/callable; order/fill arrays `(2*n,)`; event ledger. Strategic controller chooses exposure; execution chooses timing/size of child orders.
F8. Optional RL; when / interface $Q_\psi : \mathbb{R}^{d_s+2n} \to \mathbb{R}$: learned action value; $\bar{\psi}$: target parameters.	Input the full finite belief/account/mode/deadline encoding and parent action. Train from supported transitions (s, a, r_F, s') on $\mathcal{L}(\psi) = \mathbb{E}[(Q_\psi(s,a) - \{r_F - h_0\mathbf{1}_{a \ne 0} + \max_{a' \in \mathcal{A}(s')} Q_{\bar{\psi}}(s', a')\})^2]$, replacing the maximum by terminal value at the endpoint. Use RL when repeated large control solves need approximation, or action-dependent fills/information must be learned from adequate data. It is unnecessary when the fitted law and constrained MPC/Bellman solve are tractable. Validate simulator/offline coverage and preserve hard constraints; RL cannot supply an unidentified thesis edge. **Replaces F6 value solve only.**	Model callable / tensors; sufficient belief encoding (not an arbitrary posterior mean), feasible-action mask, replay transitions. Unseen actions require evidence, not extrapolation claims.
F9. Validation / gates $d_j^{\text{test}} = (\Pi_j - \Pi_j^{\text{bench}})/A_0$; J test decisions; $\widehat{\Gamma}_l$: sample lag covariance.	Chronological train/select/test blocks; test joint-path calibration, thesis persistence, catalyst forecasts, turnover, execution and net gains against frozen benchmarks. $Z = \sqrt{J}(\bar{d}^{\text{test}} - \delta_{\text{econ}})/\sqrt{\widehat{\Omega}}$, $\widehat{\Omega} = \widehat{\Gamma}_0 + 2\sum_{l=1}^{L_v}(1 - l/(L_v+1))\widehat{\Gamma}_l$. Predeclare thresholds/multiplicity; resample blocks preserving overlap and longer observed dependence. Gate solves only for valid inputs and nonempty feasible sets; otherwise funded liquidation or default/stop. Save posterior, gap, value difference and gate reason. **To next F0 and action gate.**	Scores `(J,)`, path diagnostics, block resampling; mode/risk/model/execution exits have distinct reason codes.

Composition: $O_{\le k} \to (x_k, \ell_k, y_k) \to (\widehat{\theta}, b_k, d_k, z_k) \to \mathsf{P}_k^H \to (Q_k^H, \mathcal{A}_k) \to a_k^{\text{str}} \to$ child fills $\to W_{k+1} \to b_{k+1}$.

$$a_k^* = F(O_{\le k}, s_k^{\text{acct}}, \widehat{\theta}) = \mathcal{E}_{\text{F7}}\left[\text{Gate}_{\text{F4,F9}}\left(\arg\max_{a \in \mathcal{A}_k(s_k)}\left\{\mathbb{E}_{\mathsf{P}_k^H}[r_F(s_k, a, s') + V_{k+1}^{H-1}(s')] - h_0\mathbf{1}_{a \ne 0}\right\}\right)\right].$$

SPY options: thesis paths under P, option values under Q

Baseline: one-period Heston/utility MPC. Extension: a slow valuation anchor, a fixed economic deadline and adaptive expression.



Keep the physical thesis separate from its current instrument. Observations update beliefs; strategic decisions and child-order execution use separate clocks.

Three clocks. Observations at o_j may be intraday; strategic decisions at t_k may be daily; the economic horizon ends at a fixed T^{th}, e.g. four weeks. H is the number of planned decision intervals, with $t_{k+H} \le T^{\text{th}}$. Released information is $\mathcal{F}_k$. Freeze priors, proxy definitions, calendar/surprise laws, fits, bounds, simulation count P, seed, numerical grids and test blocks. Between decisions observe/filter; only scheduled decisions or mandatory risk/exercise events authorize portfolio changes.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
O0. Released data / features $S, Y^v > 0$; J listed legs with strike K_j, expiry T_j, multiplier m_j, sign $\chi_j = \pm 1$. $C, \omega \in \mathbb{R}^J$, $X \in \mathbb{R}^{2+2J}$; economic proxy $y \in \mathbb{R}$.	Synchronize spot, bid/ask, dividends, funding, liquidity and notices; reject stale/crossed quotes. For an interval of Δ years set $Y^v = \max\{\epsilon_v, \sum_l (\Delta_l \log S)^2/\Delta\}$. Dollar midpoint $C_j = m_j(\text{bid}_j + \text{ask}_j)/2$; $\omega_j = \max\{\omega_{\min}, m_j(\text{ask}_j - \text{bid}_j)/2\}$, positive floors. $X = (S, \log Y^v, C, \omega)$, plus mask/calendar. Map released economic evidence to a predeclared log-valuation proxy y; record its formula and timestamps. **To O1–O4.**	Records; NumPy `(J,)`. Optional option grid `(N_K,N_T,2)` plus mask. Economic evidence is data, not a retrospectively revised story.
O1. Fast market / slow thesis $Z = (S, v, u) \in \mathbb{R}^2_+ \times \mathbb{R}$; u: latent log-valuation anchor; $d = u - \log S$: convergence gap. $c \in \mathbb{R}^P$: catalyst surprises. $\theta \in \Theta \subset \mathbb{R}^{12+P}$: seven Heston parameters plus slow coefficients.	Between decision dates, freeze u_k and use $\mathrm{d}S/S = \mu_k^P(S, u_k)\,\mathrm{d}t + \sqrt{v}\,\mathrm{d}W_1^P$, $\mathrm{d}v = \kappa_P(\bar{v}_P - v)\,\mathrm{d}t + \xi\sqrt{v}(\rho\,\mathrm{d}W_1^P + \sqrt{1-\rho^2}\,\mathrm{d}W_2^P)$, $\mu_k^P = \mu + \kappa_d(u_k - \log S)$. Slow model: $u_{k+1} = \bar{u} + \phi(u_k - \bar{u}) + \beta^\top c_{k+1} + \sigma_u \eta$, $y = u + \tau_y \nu$, $\beta \in \mathbb{R}^P$. Independent $\eta, \nu \sim \mathcal{N}(0,1)$; fit $0 < \phi \le 1$, $\sigma_u, \tau_y > 0$. At announced releases $c \sim \mathcal{N}(\bar{c}, C_c)$, fitted to past surprises; otherwise $c = 0$. Under $\mathbb{Q}$, stock drift is r, variance drift $\kappa_Q(\bar{v}_Q - v)$; ξ, ρ are shared. Risk prices: $\lambda_1 = (\mu_k^P - r)/\sqrt{v}$, $\xi\sqrt{v}(\rho\lambda_1 + \sqrt{1-\rho^2}\lambda_2) = \kappa_P(\bar{v}_P - v) - \kappa_Q(\bar{v}_Q - v)$. Impose $2\kappa_P\bar{v}_P, 2\kappa_Q\bar{v}_Q \ge \xi^2$, $\lvert\rho\rvert < 1$, positive variance coefficients and admissible moment/martingale bounds. Apply announced cash-dividend jumps on the stated positive-price domain. $\kappa_d = 0$ recovers constant drift. **To O2–O3.**	Fast `(2,)` / slow scalar; parameter vector `(12+p,)`. Conditional transitions are callables. Only $\mathbb{P}$ forecasts thesis convergence; the anchor is not a $\mathbb{Q}$ stock price.
O2. Filter / estimate $b_k \in \mathcal{P}(\mathbb{R}_+ \times \mathbb{R} \times \Theta)$: posterior of (v, u, θ). $\widehat{\theta}, \widehat{v}, \widehat{u}$: mode; ℓ_k: measurement likelihood.	Start with a fixed proper prior. Measurements: $\log Y^v \mid v \sim \mathcal{N}(\log v, \sigma_v^2)$, $y \mid u \sim \mathcal{N}(u, \tau_y^2)$, $C_j \mid Z, \theta \sim \mathcal{N}(c_j^A, \omega_j^2)$, conditionally independent; omit missing observations. With the O1 joint transition, $b_{k+1}(v,u,\theta) \propto \ell_{k+1}(v,u,\theta) \iint p_\theta^P(S_{k+1}, v, u \mid S_k, v', u') b_k(v', u', \theta)\,\mathrm{d}v'\,\mathrm{d}u'$. Normalize; use quadrature/particles and update released measurements between decisions. Parameters remain fixed along each future path; decisions integrate the posterior, not just its mode. Mode Greeks are a local constraint approximation. **To O3–O5.**	Weighted particles `(N_particles,14+p)` or quadrature; fit by likelihood/Bayes update. Observation noise $\sigma_v > 0$; retain posterior uncertainty.
O3. Path forecast then reprice P_k^H: physical law on $Z_{k:k+H}$. $c_j^A \in \mathbb{R}_+$: dollars/contract; $G \in \mathbb{R}^{3\times J}$: delta/gamma/vega. $g_j(S) = [\chi_j(S - K_j)]_+$.	$\mathsf{P}_k^H = \int \text{Law}_\theta^P(Z_{k:k+H} \mid S_k, v, u) b_k(\mathrm{d}v, \mathrm{d}u, \mathrm{d}\theta)$. $\mathbb{P}(Z_{k+H} \mid \mathcal{F}_k)$ is only its endpoint marginal. Draw real-world *paths*, catalyst outcomes and future noisy observations. At *each* simulated state $Z_l^{(p)}$, mark each live option with the paired $\mathbb{Q}$ Snell solver: $c_j^A(l, S, v) = \max\{m_j g_j(S), e^{-\int_l^{l'} r\,\mathrm{d}t} \mathbb{E}_\theta^Q[c_j^A(l', S', v') \mid S, v]\}$, terminal $m_j g_j$. Refine the exercise grid (including dividends/expiry) to a fixed tolerance. $G_{\cdot j} = (\partial_S c_j^A, \partial_{SS} c_j^A, 2\sqrt{v}\partial_v c_j^A)^\top$. Thus $\mathbb{P}$ weights future outcomes; $\mathbb{Q}$ gives conditional option marks, never probabilities of thesis success. Re-filter synthetic observations before each future action. **To O5–O7.**	Paths `(P,H+1,3)` (subdivide for diffusion/events); marks `(P,H+1,J)`, Greeks `(3,J)`; price/filter callables.
O4. Thesis mode / instruments $p_k = b_k(d_k > \delta)$; $z_k \in \{0, 1, \dagger\}$: waiting, active, closed; $p_{\text{on}} > p_{\text{off}}$. $E_k \in \{0,1\}^{m\times J}$: dated leg selector; chosen $C \in \mathbb{R}^m$, $G \in \mathbb{R}^{3\times m}$.	Activate at $p \ge p_{\text{on}}$; retain between thresholds. Exit on convergence $\mathbb{E}d \le \delta_{\text{exit}}$, $p \le p_{\text{off}}$, released adverse evidence $y < y_{\min}$ or $c_j < \underline{c}_j$, or fixed T^{th}; freeze thresholds and distinguish reasons. Waiting forbids new thesis risk. Closed † cancels entries and flattens thesis exposure, retaining delivery hedges; it is absorbing until a new record. Select live legs by quotes, tenor, liquidity/spread caps and delivery terms; retain held legs for exit. Use $EC, GE^\top$. Recompare *shares, calls, spreads, hedges and zero*. Expiry must cover holding or have a roll/expiry branch. An active thesis may hold zero. Baseline disables this gate and sets $\kappa_d = 0$. **To O5–O6.**	Probability/mode/deadline scalars; indices `(m,)`, selected arrays. Contract expiry, planning horizon and thesis deadline are distinct.

SPY options: dynamic policy and a multi-week example

The action is an instrument/hedge/exercise decision conditioned on a thesis belief. Assignment and future price paths remain random.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
O5. Baseline and longer control Account $s^{\text{acct}} = (q, h, B, \mathfrak{p})$, $q \in \mathbb{Z}^m, h \in l_h\mathbb{Z}, B \in \mathbb{R}$; $\mathfrak{p}$: pending claims. $a = (\Delta q, \Delta h, e^+)$: bounded integer/lot trades and owned exercise. $s = (b, z, T^{\text{th}} - t, s^{\text{acct}}, \text{pending orders})$.	Baseline $H = 1$: $a^* = \arg\max_{a \in \mathcal{A}(s)} P^{-1} \sum_p U(W_{k+1}^{\text{liq},(p)}(a))$, $U(w) = w - \lambda_U\sqrt{a_U^2 + w^2}$, $a_U > 0, 0 < \lambda_U < 1$. The ledger below generates each wealth sample. For $H > 1$, $Q_k^h(s,a) = \mathbb{E}_k^P[-\eta_T\|D_T a_{\text{trade}}\|_1 + V_{k+1}^{h-1}(s')]$, $V_k^h(s) = \max_{a \in \mathcal{A}(s)}[Q_k^h(s,a) - h_0 \mathbf{1}_{a_{\text{trade}} \neq 0}]$. At the deadline set $V^0 = U(W^{\text{liq}})$; earlier boundaries use this liquidation approximation or an estimated continuation value. One-period baseline has $\eta_T = h_0 = 0$. Expectations also average assignment/fills. Stochastic MPC executes the first action of a nonanticipative tree; full Bellman control plans to the deadline. Future actions see observations/posteriors, not latent states or later shocks. **To O6–O7.**	Finite action enumeration or global optimizer; CVXPY arrays for convex subproblems. D_T is a fixed positive diagonal trade-unit scale; $\eta_T, h_0 \geq 0$ are noncash preferences.
O5a. Risk / no-action region $\mathcal{A}(s)$: feasible controls; $b_G \in \mathbb{R}_+^3$, $\bar{q} \in \mathbb{Z}_+^m$. $\Pi_p = W_p'^{\text{liq}} - R_f W$, $R_f = e^{\int r\,\mathrm{d}t}$; $\alpha \in (0,1)$.	At every node/interim event enforce $\lvert q\rvert \leq \bar{q}$, $0 \leq e^+ \leq (q + \Delta q)_+$, legal exercise dates, $\lvert Gq + (h + h^{\text{pend}}, 0, 0)^\top\rvert \leq b_G$, and available cash minus broker margin $\geq R_{\text{cash}}$, including joint-assignment stresses. One-step tail constraint: $\widehat{\text{ES}}_\alpha(-\Pi) = \min_\zeta\{\zeta + [P(1-\alpha)]^{-1}\sum_p[-\Pi_p - \zeta]_+\} \leq L_{\text{ES}}$. If hold is feasible, the full no-action region is $\max_{a \neq 0}[Q^H(s,a) - h_0\mathbf{1}_{a_{\text{trade}} \neq 0}] \leq Q^H(s,0)$, also comparing owned exercise. Cash costs are already in wealth; turnover/switch penalties add inertia. Mandatory exercise/risk exits and O4 closure override waiting. Use fixed ties. **To O5 feasible actions.**	P&L samples `(P,)` per action; conditional mean/variance by sample moments; ES hinges; record value of holding and best alternative.
O6. Optimal execution is separate $y_e, v_e \in \mathbb{R}^{m+1}$: remaining parent trade and child rate; $T_e \ll \Delta_d$. $\Lambda_e \succ 0, \Omega_e \succeq 0 \in \mathbb{R}^{(m+1)\times(m+1)}$.	Given O5's trade target, minimize $\int_0^{T_e}[s_e^\top\lvert v_e\rvert + v_e^\top\Lambda_e v_e + \lambda_e y_e^\top\Omega_e y_e]\,\mathrm{d}u$, $\dot{y}_e = -v_e$, $y_e(0) = a_{\text{trade}}$, $y_e(T_e) = 0$, with lot/rate/legging/margin limits. Fit impact from fills and Ω_e from short-horizon quote moves. Replace O5's forecast cash costs by expected execution costs; its risk penalty is not cash. Immediate full fills remain baseline. Pending/unfilled legs return to strategic state; owned-exercise instructions bypass this trade scheduler. **To O7 fills/exercise.**	Execution optimizer/callable, arrays `(m+1,)`, order book and fill records. Allocation selects exposure; execution schedules it.
O7. Cash, delivery and assignment $K \geq 0$: actual trade cost; $e = e^+ - A \in \mathbb{Z}^m$. A_j: assigned shorts; $\beta_A, f_j \in \mathbb{R}^5$. $B^{\text{pend}}, h^{\text{pend}} \in \mathbb{R}$: unsettled cash/shares.	For actual fills: $B' = B - C^\top\Delta q - S\Delta h - K$, $q' = q + \Delta q, h' = h + \Delta h$. On exercise, $q'' = q' - e$, $\Delta h^{\text{pend}} = \sum m_j\chi_j e_j$, $\Delta B^{\text{pend}} = -\sum m_j K_j\chi_j e_j$. Transfer at delivery once. Pre-expiry baseline $A_j \sim \text{Binom}((-q_j')_+, \sigma(\beta_A^\top f_j))$, independent conditionally; $\sigma(x) = (1 + e^{-x})^{-1}$, $f = (1, \log(S/K), T - t, C - mg, D_{\text{next}}^{\text{div}})$. Fit bounded β_A by binomial likelihood. At expiry use the declared exercise/contrary-notice rule; remove unexercised legs. Notices are immediate; delivery follows its calendar. Cash accrues as $B \leftarrow R_\ell[B]_+ - R_b[-B]_+ - \text{Borrow} + h^{\text{ent}}D^{\text{div}}$. $W = B + B^{\text{pend}} + (h + h^{\text{pend}})S + q^\top C$. Liquidate at executable costs for W^{liq}; actual future marks replace predicted $\mathbb{Q}$ marks. Baseline $K = s^\top\lvert a_{\text{trade}}\rvert + a_{\text{trade}}^\top\Lambda a_{\text{trade}}$, $\Lambda \succeq 0$: observed spreads/fees and fitted impact, held fixed in forecasts. **To O5 and next posterior/account.**	Event ledger / pending-claim records, assignment samples `(P,H,m)`; cash/borrow/entitlement schedules as supplied functions. Only owned exercise is controlled.
O8. RL and validation $Q_\psi : \mathbb{R}^{d_s+d_a} \to \mathbb{R}$, $d_a = 2m + 1$; $\varphi_s(s) \in \mathbb{R}^{d_s}$: sufficient finite belief/account encoding.	Optional RL replaces O5's value solve: target $Y = -\eta_T\|D_T a_{\text{trade}}\|_1 - h_0\mathbf{1}_{a_{\text{trade}} \neq 0} + \max_{a' \in \mathcal{A}(s')} Q_{\bar{\psi}}(s', a')$, terminal maximum replaced by $U(W^{\text{liq}})$; loss $\mathbb{E}[(Q_\psi(s,a) - Y)^2]$. Useful for expensive repeated solves or learned action-dependent fills/information; unnecessary for tractable identified MPC. Require supported actions and validated simulator/offline data; preserve hard constraints. Seal chronological tests of path/catalyst calibration, $\mathbb{P}/\mathbb{Q}$ errors, assignment, costs, turnover and net benchmark gains. Blocks must preserve horizon overlap and longer dependence; adjust multiplicity. Gate solves only for valid/nonempty problems; otherwise funded exit or stop/default. **To control/update gate.**	Model callable / tensors; training tuples (s, a, r, s'). Audit posterior, gap, thesis mode, held-vs-switch value, instrument and exit reason. RL does not create the thesis forecast.

Worked conceptual example (hypothetical, normalized prices). Daily decisions; intraday observations; deadline day 20; a call expires day 40. Let filtered $u \sim \mathcal{N}(\log 110, 0.04^2)$, $\delta = 0$, $p_{\text{on}} = 0.90, p_{\text{off}} = 0.60$, $\delta_{\text{exit}} = 0.01$. **Entry:** at $S = 100$, $p \approx 0.99$; enter only if the call wins the $\mathbb{P}$-path/$\mathbb{Q}$-mark comparison after costs. **Noise:** an intraday dip with unchanged thesis evidence does not trigger a decision or invalidate it. **Week 2:** at $S = 106$, remaining gap $\log(110/106) = 0.037$, $p \approx 0.82$: retain the thesis. If the after-cost value gain from a smaller call spread (or shares) exceeds h_0, switch; otherwise hold. **Exit:** $S = 109$ gives gap $0.009 < \delta_{\text{exit}}$, so close on convergence. Alternatively a failed catalyst moving the anchor to 102 at $S = 106$ gives $p \approx 0.17$: invalidate. Day 20 closes any remaining thesis risk; zero exposure can be optimal earlier while the thesis remains active. **Composition:** $O \to (X, y) \to (b, \widehat{\theta}, d, z) \to \mathsf{P}_k^H \to \{c^{A,Q}(Z_l^{(p)}), W_l^{(p)}\} \to (Q^H, \mathcal{A}) \to a^{\text{str}} \to \text{fills} \to W' \to b'$.

$$a_k^* = F(O_{\leq k}, s_k^{\text{acct}}; b_k, \widehat{\theta}, \widehat{\beta}_A) = \mathcal{E}_{\text{O6}}\left[\text{Gate}_{\text{O4,O8}}\left(\arg\max_{a \in \mathcal{A}(s_k)}\left\{\mathbb{E}_{\substack{Z \sim \mathsf{P}_k^H,\ C_l = c^{A,Q}(Z_l) \\ s' = \text{Ledger}_{\text{O7}}(s_k, a)}}[-\eta_T\|D_T a_{\text{trade}}\|_1 + V_{k+1}^{H-1}(s')] - h_0\mathbf{1}_{a_{\text{trade}} \neq 0}\right\}\right)\right].$$